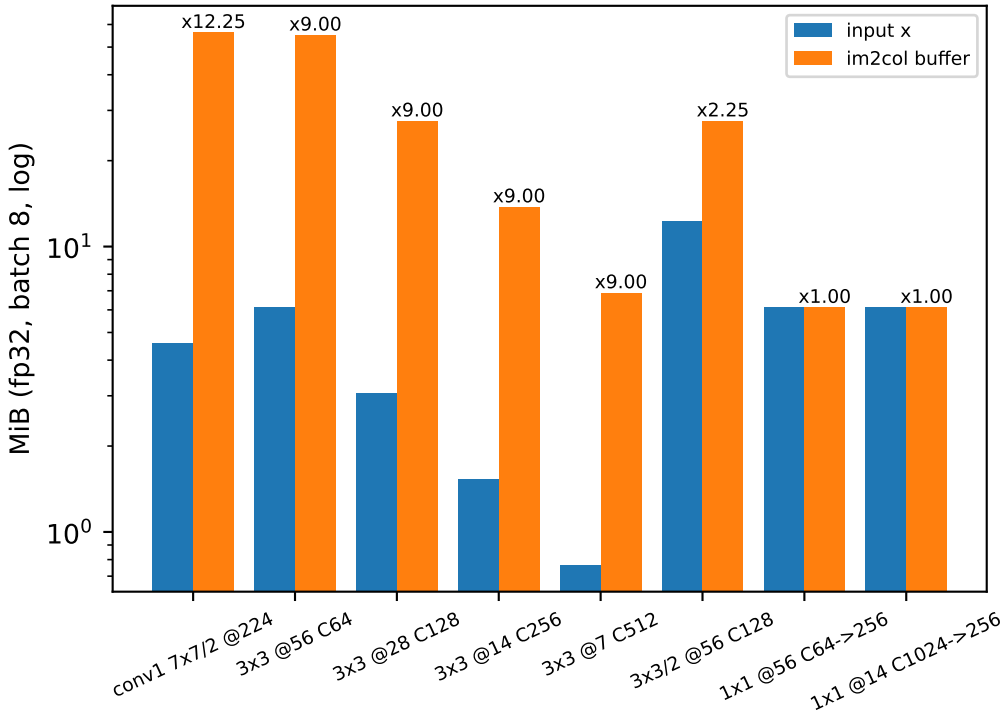
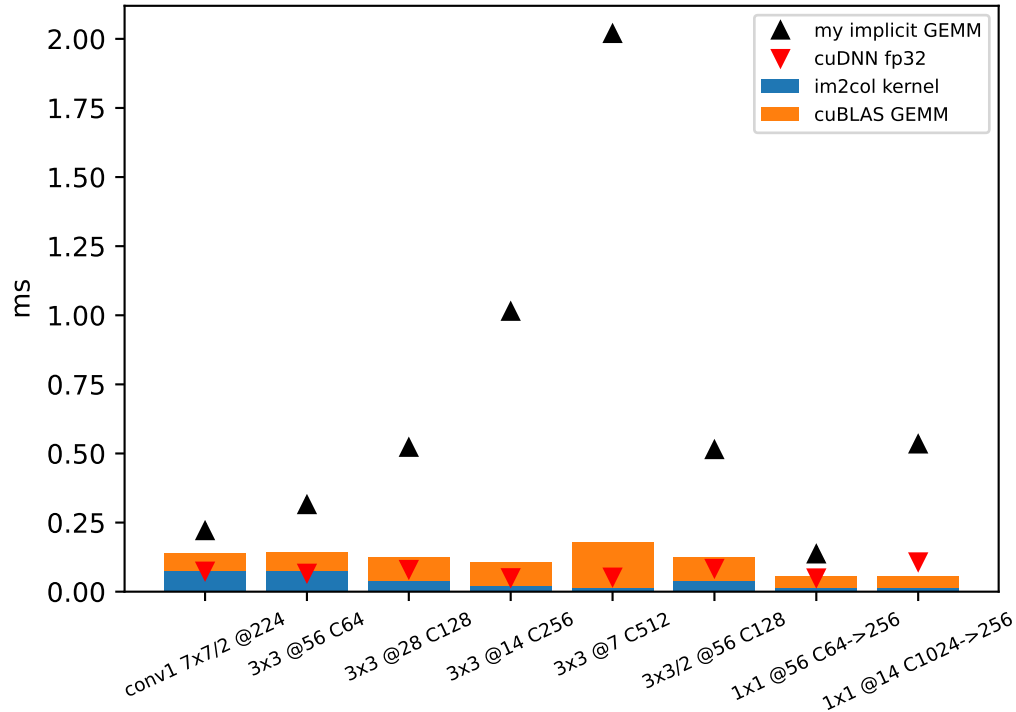

im2col memory blow-up on GPU
MiB (fp32, batch 8, log)
10^1
10^0
input x
im2col buffer
x12.25
x9.00
x9.00
x9.00
x9.00
x2.25
x1.00
x1.00
conv1 7x7/2 @224
3x3 @56 C64
3x3 @28 C128
3x3 @14 C256
3x3 @7 C512
3x3/2 @56 C128
1x1 @56 C64->256
1x1 @14 C1024->256


Explicit im2col: copy vs GEMM time
ms
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
my implicit GEMM
cuDNN fp32
im2col kernel
cuBLAS GEMM
conv1 7x7/2 @224
3x3 @56 C64
3x3 @28 C128
3x3 @14 C256
3x3 @7 C512
3x3/2 @56 C128
1x1 @56 C64->256
1x1 @14 C1024->256